TAKAHIRO MAEDA

Machine Learning Researcher at Toshiba Corporation
◇ wtqkqlzlz@gmail.com ◇ https://meaten.github.io ◇ Last Update July 29, 2026

HIGHLIGHTS

- Strong publications including first-authored CVPR, ICCV, RA-L papers.
- Expertise in enterprise-tuned VLMs, flow models, and safety-critical prediction systems.
- Worked at Toyota and CMU. Won 1st place in MPII Real Robot Challenge (Phases 2 & 3).

WORK EXPERIENCE

Toshiba Corporation, Japan — Research and Development Center *April 2025 - Present*
Research Engineer

- Prototyping domain-adapted VLMs for enterprise document understanding, focusing on model robustness and handling domain-specific terminology.
- Developing a context-engineering framework for AI agents to more reliably extract domain knowledge and perform tool-using tasks.
- Created an initial internal benchmark for technical drawing interpretation to standardize evaluation of multimodal reasoning models.
- Collaborating with business units to identify workflow automation opportunities and define requirements for future AI-agent integrations.

EDUCATION

Toyota Technological Institute, Japan *November 2021 - March 2025*
PhD in Engineering, Electronics and Information Engineering

Toyota Technological Institute, Japan *April 2019 - March 2021*
Master of Engineering, Electronics and Information Engineering

Toyota Technological Institute, Japan *April 2015 - March 2019*
Bachelor of Engineering

RESEARCH INTEREST

Computer vision, robotics, human motion prediction, and multimodal models. Currently developing enterprise VLMs and AI agents at Toshiba, with long-term interests in robust perception for intelligent assistive robots.

PUBLICATIONS

T. Maeda, J. Cao, N. Ukita, and K. Kitani.
CacheFlow: Fast Human Motion Prediction by Cached Normalizing Flow *Under Review*

H. Taketsugu, T. Oba, **T. Maeda**, S. Nobuhara, and N. Ukita.
Physical plausibility-aware trajectory prediction via locomotion embodiment *CVPR, 2025*

T. Maeda, K.Takeshita, N. Ukita, and K. Tanaka.
Multimodal Active Measurement for Human Mesh Recovery in Close Proximity *RA-L, 2024*

T. Maeda and N. Ukita.
Fast Inference and Update of Probabilistic Density Estimation on Trajectory Prediction *ICCV, 2023*

T. Maeda and N. Ukita.
MotionAug: Augmentation with Physical Correction for Human Motion Prediction. *CVPR, 2022*

T. Yoneda, C.Schaff, **T. Maeda**, & M. Walter.
Grasp and motion planning for dexterous manipulation for the real robot challenge. *ArXiv, 2021*

T. Matsumoto, K. Shimosato, **T. Maeda**, T. Murakami, K. Murakoso, K. Mino, & N. Ukita.
Automatic Human Pose Annotation For Loose-Fitting Clothes. *MVA, 2019*

SKILLS

Pytorch, TensorFlow	
LLMs,VLMs	
Prompt/Context Automatic Optimization	
Model serving	vLLM, Ollama, HF
Agent Architecture	ReAct, Tool Use, Computer Use
Agent Frameworks	LangChain, Google ADK, OpenHands
Generative Models	Diffusion, Flow Matching
Robotics	ROS, Toyota Human Support Robot
English	TOEFL iBT 94 (R:29 L:24 S:18 W:23, July 2024)

LICENSES

Certified Care Worker（介護福祉士国家資格） April 2022, Japan

EXPERIENCES

Robotics Instutute, Carnegie Mellon University — Research Internship *Sep-Dec 2024*

Frontier Research Center, Toyota Motor Corporation — Research Internship *May-Aug 2023*

Toyota Technological Institute at Chicago — Research Internship *Sep-Dec 2019*

SERVICES

Reviewer for CVPR, ICCV, NeurIPS, TPAMI, CVIU.

HONORS & AWARDS

Conference Presentation Award
IEEE Japan Nagoya Section, 2023

1st Place (Phase 2, €3500) & **1st Place** (Phase 3, €9000)
Real Robot Challenge 2021, the Max Planck Institute for Intelligent Systems

Best Poster Award
Machine Vision and Applications Organization, Tokyo, 2019